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Power supply device, and vehicle and power storage device each equipped with same

Abstract

Power supply device includes: a battery stack in which a plurality of battery cells each having an electrode terminal on an upper surface of an outer covering can are stacked; and a plurality of bus bars connecting electrode terminals of adjacent battery cells to each other. Bus bar includes window opened through which a part of electrode terminal of battery cell can be visually recognized in a state of being overlapped with electrode terminal in plan view. As a result, even when bus bar is overlapped on electrode terminal, electrode terminal on the lower side can be visually recognized through window, so that the connection position and the height can be confirmed, and the reliability of connection can be enhanced.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a U.S. national stage application of the PCT International Application No. PCT/JP2020/043106 filed on Nov. 19, 2020, which claims the benefit of foreign priority of Japanese patent application No. 2020-064058 filed on Mar. 31, 2020, the contents all of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a power supply device, a vehicle provided with the same, and a power storage device.

BACKGROUND ART

(3) A power supply device such as a battery module or a battery pack including a plurality of battery cells is used as a power source for a vehicle such as a hybrid vehicle or an electric vehicle, a power source for a power storage system for a factory, a home, or the like (See, for example, PTLs 1 to 3).

(4) In such a power supply device, a plurality of chargeable and rechargeable battery cells are stacked. For example, as shown in a schematic sectional view of FIG. **16**, power supply device **900** has end plates **903** on both end surfaces of battery stack **910** obtained by stacking battery cells **901** having prismatic outer covering can. End plates **903** are fastened to each other with bind bars **904**. Battery cell **901** having a square shape is provided with, on the upper surface, positive and negative electrode terminals **902** separated from each other. Electrode terminals **902** of adjacent battery cells **901** are connected by bus bars **940** as shown in the enlarged exploded perspective view of FIG. **17**.

(5) However, as illustrated in the exploded perspective view of FIG. **18** and the plan view of FIG. **19**, when bus bar **840** is larger than electrode terminal **802**, there is a problem that electrode terminal **802** is hidden by bus bar **840** and the welding position cannot be seen.

(6) In the case of laser welding, if there is a gap between parts to be welded, the strength of the welding is not stabilized and the reliability is affected. Therefore, it is necessary to perform laser welding without a gap. However, when electrode terminal **802** on the lower side is hidden by bus bar **840** as illustrated in FIG. **19**, the gap therebetween cannot be confirmed.

CITATION LIST

Patent Literature

(7) PTL 1: Unexamined Japanese Patent Publication No. 6299537 PTL 2: Unexamined Japanese Patent Publication No. 5940374 PTL 3: WO 2019-159732 A

SUMMARY OF THE INVENTION

Technical Problem

(8) An object of an aspect of the present invention is to provide a power supply device with improved reliability of connection between an electrode terminal of a battery cell and a bus bar, and a vehicle and a power storage device including the power supply device.

Solution to Problem

(9) A power supply device according to an aspect of the present invention includes: a battery stack in which a plurality of battery cells each including an electrode terminal on an upper surface of an outer covering can are stacked; and a plurality of bus bars that connect electrode terminals of

adjacent battery cells to each other, in which each of the bus bars includes a window through which a part of the electrode terminal can be visually recognized in a state of overlapping the electrode terminal of the battery cells in plan view.
Advantageous Effect of Invention
(10) According to a power supply device according to an aspect of the present invention, even when a bus bar is overlapped on an electrode terminal, a lower electrode terminal can be visually recognized through a window, so that a connection position and a height can be confirmed, and reliability of connection can be enhanced.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a perspective view with an enlarged view of a main part illustrating a power supply device according to a first exemplary embodiment of the present invention.
(2) FIG. **2** is an exploded perspective view with an enlarged view of a main part of the power supply device illustrated in FIG. **1**.
(3) FIG. **3** is a schematic cross-sectional view illustrating battery cells of a power supply device according to a modified example.
(4) FIG. **4** is a schematic plan view illustrating a connecting piece of a bus bar.
(5) FIG. **5** is a schematic cross-sectional view taken along line V-V in FIG. **4**.
(6) FIG. **6** is a schematic plan view illustrating a connecting piece of a bus bar of a power supply device according to a second exemplary embodiment.
(7) FIG. **7** is a schematic plan view illustrating a connecting piece of a bus bar of a power supply device according to a third exemplary embodiment.
(8) FIG. **8** is a schematic plan view illustrating a connecting piece of a bus bar of a power supply device according to a fourth exemplary embodiment.
(9) FIG. **9** is a schematic plan view illustrating a connecting piece of a bus bar of a power supply device according to a fifth exemplary embodiment.
(10) FIG. **10** is a schematic plan view illustrating a connecting piece of a bus bar of a power supply device according to a sixth exemplary embodiment.
(11) FIG. **11** is a schematic plan view illustrating a connecting piece of a bus bar of a power supply device according to a seventh exemplary embodiment.
(12) FIG. **12** is a schematic plan view illustrating a connecting piece of a bus bar of a power supply device according to an eighth exemplary embodiment.
(13) FIG. **13** is a block diagram illustrating an example in which a power supply device is mounted on a hybrid vehicle that travels with an engine and a motor.
(14) FIG. **14** is a block diagram illustrating an example in which a power supply device is mounted on an electric automobile that travels only with a motor.
(15) FIG. **15** is a block diagram illustrating an example of applying a power supply device for power storage.
(16) FIG. **16** is an exploded perspective view of a conventional power supply device.
(17) FIG. **17** is an exploded perspective view illustrating bus bars and electrode terminals in FIG. **16**.
(18) FIG. **18** is an exploded perspective view of a bus bar and electrode terminals.
(19) FIG. **19** is a schematic plan view illustrating a state in which the bus bar of FIG. **18** is overlapped on the electrode terminals.
DESCRIPTION OF EMBODIMENT
(20) Exemplary embodiments of the present invention may be specified by the following configurations.

(21) In a power supply device according to an exemplary embodiment of the present invention, in addition to the above configuration, the bus bar has a plurality of windows opened, and the electrode terminal can be visually recognized from the windows. With the above configuration, since different positions of the electrode terminal can be confirmed through the plurality of windows, the height of different parts of the electrode terminal can be measured, and the gap between the back surface of the bus bar and the electrode terminal can be calculated from the known thickness of the bus bar.
(22) In a power supply device according to another exemplary embodiment of the present invention, in addition to any one of the above configurations, the bus bar includes the plurality of windows opened while separating the plurality of windows so as to sandwich a fixing region where the bus bar and the electrode terminal are connected. With the above configuration, since the height of the electrode terminal can be measured at two positions facing each other with the fixing region interposed therebetween, it is possible to more accurately calculate the gap between the electrode terminal and the bus bar in the fixing region.
(23) In addition, in a power supply device according to another exemplary embodiment of the present invention, in addition to any of the configurations described above, the plurality of windows are opened at diagonal positions of the electrode terminal. With the above configuration, since the height of the electrode terminal can be measured at two positions in a diagonal line shape, it is possible to more accurately calculate the gap between the electrode terminal and the bus bar.
(24) Furthermore, in a power supply device according to another exemplary embodiment of the present invention, in addition to any of the configurations described above, the windows includes an outer-shape window for detecting an outer shape of the electrode terminal, and a height window for detecting a height of the electrode terminal. With the above configuration, by preparing each of the outer-shape window for detecting the outer shape of the electrode terminal and the height window for detecting the height of the electrode terminal among the plurality of windows, processing such as outer shape detection and height detection can be performed in each window, and by providing the window at a position suitable for outer shape detection and a position suitable for height detection, an appropriate processing result can be obtained.
(25) Furthermore, in a power supply device according to another exemplary embodiment of the present invention, in addition to any of the configurations described above, the window is opened in a rectangular shape in the bus bar.
(26) Furthermore, in a power supply device according to another exemplary embodiment of the present invention, in addition to any of the configurations described above, the window is opened in an elongated hole shape longer than a width of the electrode terminal. With the above configuration, the facing end edges of the electrode terminal can be visually recognized from one window, and the inclination of the electrode terminal can be detected from one window.
(27) Furthermore, in a power supply device according to another exemplary embodiment of the present invention, in addition to any of the configurations described above, the window is formed to be smaller than the electrode terminal. With the above configuration, it is possible to grasp and fix the position of the electrode terminal with the window through which only a part of the electrode terminal can be visually recognized without providing the bus bar with a round hole matching the projecting strip of the electrode terminal as in the conventional case.
(28) Furthermore, in a power supply device according to another exemplary embodiment of the present invention, in addition to any of the configurations described above, a connecting part between the bus bar and the electrode terminal is a laser welding part.
(29) Furthermore, in a power supply device according to another exemplary embodiment of the present invention, in addition to any of the configurations described above, the electrode terminal forms a recess. With the above configuration, it is possible to perform joining without using conventional engagement between the electrode terminal including a protrusion and the round hole of the bus bar.

(30) Furthermore, in a power supply device according to another exemplary embodiment of the present invention, in addition to any of the configurations described above, the bus bar is formed to be larger than the electrode terminal of the battery cell in a state of overlapping the electrode terminal in plan view. With the above configuration, even when the bus bar is larger than the electrode terminal and the electrode terminal is inevitably hidden by the bus bar, the electrode terminal on the lower side can be visually recognized through the window, so that the welding position and the height can be confirmed, and the reliability of laser welding can be enhanced.
(31) Further, an electric vehicle according to another exemplary embodiment of the present invention includes: any one of the above-mentioned power supply devices; a traveling motor to which electric power is supplied from the power supply device; a vehicle main body on which the power supply device and the motor are mounted; and wheels that are driven by the motor to make the vehicle main body travel.
(32) Further, a power storage device according to another exemplary embodiment of the present invention includes: any one of the above-mentioned power supply devices; and a power supply controller that controls charging to and discharging from the power supply device, in which the power supply controller enables charging to the battery cell with electric power from the outside and controls the battery cell to perform charging.
(33) Furthermore, a method for manufacturing a power supply device according to another exemplary embodiment of the present invention includes the steps of: preparing a battery stack in which a plurality of battery cells each including an electrode terminal are stacked, and a bus bar that connects electrode terminals of adjacent battery cells to each other, the bus bar including a window opened in advance; confirming positioning to confirm whether the bus bar and the electrode terminal are correctly positioned at a welding position by overlapping the bus bar on the electrode terminal of the battery cell, and visually recognizing a part of the electrode terminal through the window; and welding the bus bar whose positioning is confirmed to the battery cell. As a result, even when the bus bar is overlapped on the electrode terminal, the lower electrode terminal can be visually recognized through the window, so that the welding position and the height can be confirmed, and the reliability of welding can be enhanced.
(34) Furthermore, in a method for manufacturing a power supply device according to another exemplary embodiment of the present invention, in addition to the above, the step of confirming positioning includes the steps of: measuring a height of the electrode terminal by visually recognizing a part of the electrode terminal through the window in a state where the bus bar is overlapped with the electrode terminal of the battery cell; and calculating a gap between a back surface of the bus bar and a front surface of the electrode terminal based on a known thickness of the bus bar. As a result, by acquiring the gap between the back surface of the bus bar and the electrode terminal, it is possible to take necessary measures since the strength of welding between the bus bar and the electrode terminal cannot be secured when the gap is large, and the reliability of welding can be improved.
(35) Furthermore, in the method for manufacturing the power supply device according to another exemplary embodiment of the present invention, in addition to any of the above, the step of welding the bus bar to the battery cell is performed by laser welding.
(36) Hereinafter, exemplary embodiments of the present invention are described with reference to the drawings. However, the exemplary embodiments described below are examples for embodying the technical idea of the present invention, and the present invention is not limited to the exemplary embodiments described below. Further, in the present specification, members indicated in the claims are not limited to the members of the exemplary embodiments. In particular, the dimensions, materials, shapes, and the relative arrangement of the constituent members described in the exemplary embodiments are not intended to limit the scope of the present invention only thereto unless otherwise specified and are merely illustrative examples. Note that the sizes and positional relationships of the members illustrated in the drawings may be exaggerated for clarity of

description. Further, in the following description, the same names and marks indicate the same or similar members, and detailed description will be appropriately omitted. Furthermore, the elements constituting the present invention may be configured such that a plurality of elements are constituted of the same member to form one member that functions as the plurality of elements, or conversely, a function of one member can be shared and achieved by a plurality of members. In addition, the description in some examples or exemplary embodiments may be applied to other examples, exemplary embodiments, and the like.
(37) A power supply device according to the exemplary embodiment is used for various applications such as a power source that is mounted on an electric vehicle as a hybrid vehicle or an electric automobile and supplies electric power to a drive motor, a power source that stores generated electric power of natural energy such as solar power generation or wind power generation, and a power source that stores midnight electric power, and is particularly used as a power source suitable for high-power and high-current applications. In the following example, an exemplary embodiment applied to a power supply device for driving an electric vehicle will be described.
First Exemplary Embodiment
(38) Power supply device **100** according to a first exemplary embodiment of the present invention is illustrated in FIGS. **1** and **2**. In these drawings, FIG. **1** is a perspective view with an enlarged view of a main part of power supply device **100** according to a first exemplary embodiment, and FIG. **2** is an exploded perspective view with an enlarged view of the main part of power supply device **100** illustrated in FIG. **1**.
(39) Power supply device **100** shown in these drawings includes battery stack **10** in which a plurality of battery cells **1** are stacked with insulating spacers **16** interposed therebetween, a pair of end plates **20** covering both end surfaces of battery stack **10**, a plurality of fastening members **15** for fastening end plates **20** to each other, and bus bar **40** arranged on the upper surface of battery stack **10**.
(40) Each of fastening members **15** is formed in a plate shape extended in a stacking direction of the plurality of battery cells **1**. Fastening members **15** are disposed on opposite side surfaces of battery stack **10**, respectively, to fasten end plates **20** to each other.
(41) (Battery Stack **10**)
(42) As shown in FIG. **2**, battery stack **10** includes the plurality of battery cells **1** each including positive and negative electrode terminals **2**, and bus bars **40** connected to electrode terminals **2** of the plurality of battery cells **1** to connect the plurality of battery cells **1** in parallel and in series. The plurality of battery cells **1** are connected in parallel or in series through bus bars **40**. Battery cell **1** is a chargeable and dischargeable secondary battery. In power supply device **100**, the plurality of battery cells **1** are connected in parallel to form a parallel battery group, and a plurality of the parallel battery groups are connected in series, so that a large number of battery cells **1** are connected in parallel and in series. In power supply device **100** illustrated in FIG. **2**, the plurality of battery cells **1** are stacked to form battery stack **10**. Further, the pair of end plates **20** is disposed on both end surfaces of battery stack **10**. End parts of fastening members **15** are fixed to end plates **20**, and battery cells **1** in a stacked state are fixed in a pressed state.
(43) (Battery Cell **1**)
(44) As illustrated in FIG. **2**, battery cells **1** are prismatic batteries having a width larger than the thickness, in other words, a width smaller than the width, and are stacked in a thickness to form battery stack **10**. Each of battery cells **1** can be, for example, a lithium ion secondary battery. Further, the battery cell can be any chargeable secondary battery such as a nickel metal hydride battery and a nickel cadmium battery. Battery cell **1** houses positive and negative electrode plates in exterior can **1***a* having a sealed structure together with an electrolyte solution. Outer covering can **1***a* is obtained by press-molding a metal sheet such as aluminum or an aluminum alloy into a prismatic shape, and hermetically sealing an opening part with sealing plate **1***b*. Sealing plate **1***b* is

made of aluminum or aluminum alloy similarly as rectangular parallelepiped exterior can **1***a*, and positive and negative electrode terminals **2** are respectively fixed to both end parts of sealing plate **1***b*. Further, sealing plate **1***b* is provided with, between positive and negative electrode terminals **2**, gas discharge valve **1***c* as a safety valve that opens in response to a change in pressure inside each of battery cells **1**.

(45) The plurality of battery cells **1** are stacked such that the thickness direction of each battery cell **1** is the stacking direction to constitute battery stack **10**. At this time, the power of battery stack **10** can be increased by making the number of the battery cells stacked larger than usual. In such a case, battery stack **10** is long extended in the stacking direction. In battery cells **1**, terminal surfaces **1**X on which positive and negative electrode terminals **2** are provided are disposed on the same plane, and the plurality of battery cells **1** are stacked to form battery stack **10**. A top surface of battery stack **10** is a surface on which gas discharge valves **1***c* of the plurality of battery cells **1** are provided.

(46) (Electrode Terminal **2**)

(47) In each of battery cells **1**, as illustrated in FIG. **2** and the like, sealing plate **1***b* as a top surface is used as terminal surface **1**X, and positive and negative electrode terminals **2** are fixed to both ends of terminal surface **1**X. Electrode terminal **2** has a chamfered rectangular plate shape. However, the shape of the electrode terminal is not limited thereto, and may be a polygonal shape such as an octagonal shape, a circular shape, or an elliptical plate shape. It is not always necessary to chamfer the corner part of the plate-shaped electrode terminal. Alternatively, as illustrated in the schematic cross-sectional view of FIG. **3**, recess **2***a* may be formed at the center of the plate shape.

(48) Positions at which positive and negative electrode terminals **2** are fixed to sealing plate **1***b* of battery cell **1** are set to be positions where the positive electrode and the negative electrode are bilaterally symmetrical. This enables adjacent battery cells **1** to be connected in series by stacking battery cells **1** in an alternately and horizontally reversed manner and connecting electrode terminals **2** of the positive electrode and the negative electrode that are adjacent and close to each other by bus bars **40**, as shown in FIG. **2**. Note that the present invention does not specify the number and the connection state of the battery cells constituting the battery stack. The number of battery cells constituting the battery stack and the connection state of the battery cells may be modified in various manners, inclusive of other exemplary embodiments to be described later.

(49) The plurality of battery cells **1** are stacked such that the thickness direction of each battery cell **1** is the stacking direction to constitute battery stack **10**. In battery stack **10**, the plurality of battery cells **1** are stacked such that terminal surfaces **1**X on which positive and negative electrode terminals **2** are provided, that is, sealing plates **1***b* in FIG. **2** are flush with each other.

(50) (Insulating Spacer **16**)

(51) In battery stack **10**, insulating spacer **16** is interposed between battery cells **1** stacked adjacent to each other. Insulating spacer **16** is made of insulating material such as resin in the form of a thin plate or sheet. Insulating spacer **16** has a plate shape having substantially the same size as a facing surface of battery cell **1**. Insulating spacer **16** can be stacked between adjacent battery cells **1** to insulate adjacent battery cells **1** from each other. Note that, as a spacer disposed between adjacent battery cells, a spacer having a shape in which a flow path of a cooling gas is formed between the battery cell and the spacer can also be used. In addition, the surface of the battery cell can be covered with an insulating material. For example, the surface of the exterior can except for the electrode terminal portion of the battery cell may be covered with a shrink film such as a polyethylene terephthalate (PET) resin.

(52) Power supply device **100** illustrated in FIG. **2** includes end plates **20** disposed on respective opposite end surfaces of battery stack **10**. Note that end surface spacer **17** may be interposed between each of end plates **20** and battery stack **10** to insulate the end plate and the battery stack from each other. End surface spacer **17** can also be formed in the shape of a thin plate or a sheet by using an insulating material such as resin.

(53) In power supply device **100** according to the first exemplary embodiment, in battery stack **10** in which the plurality of battery cells **1** are stacked on each other, electrode terminals **2** of the plurality of battery cells **1** adjacent to each other are connected by bus bars **40** to connect the plurality of battery cells **1** in parallel and in series.
(54) (End Plate **20**)
(55) As illustrated in FIG. **2**, end plates **20** are disposed at both ends of battery stack **10**, and are fastened via a pair of left and right fastening members **15** disposed along both side surfaces of battery stack **10**. End plates **20** are disposed at both ends of battery stack **10** in the stacking direction of battery cells **1** and outside end surface spacers **17** to sandwich battery stack **10** from both ends.
(56) (Fastening Member **15**)
(57) Both ends of fastening member **15** are fixed to end plates **20** disposed on both end surfaces of battery stack **10**. End plates **20** are fixed by a plurality of fastening members **15**, whereby battery stack **10** is fastened in the stacking direction. As illustrated in FIG. **2** and the like, fastening members **15** are each made of metal having a predetermined width and a predetermined thickness along the side surface of battery stack **10**, and are disposed to face both side surfaces of battery stack **10**. As each of fastening members **15**, a metal sheet such as iron, preferably a steel plate can be used. Fastening member **15** made of a metal sheet is bent by press molding or the like to be formed into a predetermined shape.
(58) Fastening member **15** is bent in a U-shape above and below plate-shaped fastening main surface **15***a* to form bent pieces **15***d*. Upper and lower bent pieces **15***d* cover upper and lower surfaces of battery stack **10** from corners on left and right side surfaces of battery stack **10**. Fastening member **15** is fixed to outer peripheral surfaces of end plates **20** by screwing bolts **15***f* into a plurality of fastening screw holes opened in fastening main surface **15***a*. Note that the fixing between fastening main surface **15***a* and each of end plates **20** is not necessarily limited to the screwing using a bolt, and may be a pin, a rivet, or the like.
(59) Power supply device **100** in which a large number of battery cells **1** are stacked is configured to bind the plurality of battery cells **1** by connecting end plates **20** disposed at both ends of battery stack **10** including the plurality of battery cells **1** by fastening members **15**. By binding the plurality of battery cells **1** via end plates **20** and fastening members **15** having high rigidity, it is possible to suppress expansion, deformation, relative movement, malfunction due to vibration, and the like of battery cells **1** due to charging and discharging, and degradation.
(60) (Insulating Sheet **30**)
(61) Further, insulating sheet **30** is interposed between each of fastening members **15** and battery stack **10**. Insulating sheet **30** is made of material having an insulating property, such as resin, and insulates metal fastening member **15** from battery cell **1**. Insulating sheet **30** shown in FIG. **2** and the like is constituted of flat plate **31** for covering the side surface of battery stack **10**, and bent cover parts **32** respectively provided on an upper part and a lower part of flat plate **31**. Bent cover parts **32** are bent in a U shape from flat plate **31** so as to cover bent pieces **15***d* of fastening member **15**, and are then further folded back. Accordingly, since bent pieces **15***d* are covered with the insulating bent cover parts from the top surface to the side surface and the lower surface, unintended conduction between battery cells **1** and fastening members **15** can be avoided.
(62) Further, bent pieces **15***d* press the top surface and the lower surface of each of battery cells **1** of battery stack **10** via bent cover parts **32**. Consequently, each battery cell **1** is held in a height direction by being pressed by bent pieces **15***d* in a up-down direction, and even if vibration, impact, or the like is applied to battery stack **10**, each battery cell **1** can be maintained so as not to be displaced in the up-down direction.
(63) Note that, when the battery stack or the surface of the battery stack is insulated, for example, when the battery cell is housed in an insulating case, or covered with a heat-shrinkable film made of resin, or when an insulating paint or coating is applied to the surfaces of the fastening members,

or when the fastening members are made of an insulating material, the insulating sheet can be unnecessary. In addition, regarding insulating sheet **30**, bent cover part **32** may be provided only on the upper end in the case where insulation from bent piece **15***d* of fastening member **15** does not need to be taken into consideration near the lower surface of battery stack **10**. This corresponds to, for example, the case where the battery cell is covered with a heat-shrinkable film.
(64) (Bus Bar **40**)
(65) Both end parts of bus bar **40** are connected to positive and negative electrode terminals **2** to connect battery cells **1** in series or in parallel. In power supply device **100**, battery cells **1** are connected in series to increase the output voltage, and battery cells **1** are connected in series and in parallel to increase the output voltage and the output current.
(66) As the connection between bus bar **40** and electrode terminal **2**, for example, welding can be used. Laser beam scanning can be used for welding. In particular, an inexpensive and high-output fiber laser can be suitably used. Alternatively, resistance welding may be used instead of laser welding. In the laser welding, bus bar **40** and electrode terminal **2** are partially melted and fixed by scanning a laser beam several times. Such a region where bus bar **40** and electrode terminal **2** are fixed is referred to as fixing region FA in the present specification. In the case of fixing by laser welding, fixing region FA is scanned with a laser beam to fix bus bar **40** and electrode terminal **2**.
(67) (Connecting Piece **41**)
(68) Bus bar **40** shown in FIGS. **1** and **2** connects opposed electrode terminals **2** of two adjacent battery cells **1**. Therefore, bus bar **40** is extended so as to straddle electrode terminals **2** of adjacent battery cells **1**, and covers electrode terminals **2**. In other words, each bus bar **40** is formed by connecting two connecting pieces **41** connecting electrode terminals **2**, and each connecting piece **41** is connected to electrode terminals **2** of different battery cells **1** to electrically connect these battery cells **1** to each other. That is, two electrode terminals **2** are arranged so as to be covered with one bus bar **40** from the upper surface. In the example of FIG. **1** and the like, connecting pieces **41** constitutes one bus bar **40** by connecting two connecting pieces **41**, but the number of connecting pieces is not limited to two, and three or more connecting pieces may be connected to constitute a bus bar according to the number of connection and the connection form of the battery cells.
(69) (Window **42**)
(70) As illustrated in the plan view of FIG. **4**, window **42** is opened in each connecting piece **41** of bus bar **40**. By opening window **42**, a part of electrode terminal **2** can be visually recognized in a state where bus bar **40** overlaps electrode terminal **2** of battery cell **1**. In this way, even if bus bar **40** is overlapped from above electrode terminal **2**, electrode terminal **2** on the lower side can be visually recognized through window **42**, the welding position and the height can be confirmed, and the reliability of the laser welding can be enhanced.
(71) In the conventional power supply device, when the bus bar is overlapped on the upper surface of the electrode terminal at the time of welding the bus bar and the electrode terminal, there is a problem that the electrode terminal is hidden by the bus bar and the position of the electrode terminal cannot be visually recognized. For example, even if the position of the electrode terminal hidden by the bus bar is deviated from the assumed position, it cannot be confirmed from the bus bar, so that it has been considered that laser welding is not appropriately performed and a predetermined welding strength cannot be exhibited.
(72) Therefore, in the present exemplary embodiment, window **42** is provided in bus bar **40** so that a part of electrode terminal **2** can be visually recognized, thereby improving the estimation accuracy of the welding position. In particular, by setting window **42** in advance at a position where the outer shape of electrode terminal **2** assumed as a welding position is accommodated, it is possible to confirm that electrode terminal **2** is correctly arranged.
(73) As another point of view, in the conventional power supply device, as illustrated in the exploded perspective view of FIG. **16** and the enlarged perspective view of FIG. **17**, a protrusion is

formed at the central part of electrode terminal **902** formed on the upper surface of the outer covering can of battery cell **901**, a round hole into which the protrusion is inserted is formed in one bus bar **940**, and the protrusion is inserted into the round hole to perform laser welding. With this configuration, when bus bar **940** and electrode terminal **902** are welded, positioning can be performed with the protrusion and the round hole, and the welding position can also be visually recognized. However, this configuration has a problem that the welding strength cannot be increased in proportion to the area because the contact area is small at the time of welding with bus bar **940** and the area that can be welded is limited.
(74) On the other hand, it is conceivable that a protrusion whose area is locally narrowed is not formed at the central part of the electrode terminal of the battery cell but is formed as a flat surface, and one bus bar is also formed as a flat surface to increase the contact area thereof to gain the area of welding. However, in this configuration, since there is no round hole for inserting the protrusion on the bus bar, there is a new problem that the outer shape of the electrode terminal cannot be visually recognized. In particular, when the electrode terminal is smaller than the connecting piece of the bus bar, the welding position is hidden by the bus bar, so that it is not possible to confirm whether the electrode terminal and the bus bar are accurately positioned. If the laser welding is performed in this state, the joint strength is not guaranteed for the electrode terminal and the bus bar which are not correctly positioned, and there is a possibility that the reliability is lowered.
(75) Therefore, in the present exemplary embodiment, as described above, by opening positioning window **42** in bus bar **40** instead of the engagement with the electrode terminal, the position of electrode terminal **2** can be confirmed, and it can be confirmed that the relative positioning between electrode terminal **2** and bus bar **40** is correctly performed. Window **42** is opened in advance at a position where a part of the outer shape of electrode terminal **2** can be visually recognized in a case where it is assumed that electrode terminal **2** is at a predetermined position assumed as a fixing position in a state where bus bar **40** is overlapped with the fixing position at which electrode terminal **2** is joined. By opening window **42** at such a position, it is possible to confirm through window **42** that electrode terminal **2** is correctly arranged at an assumed position on the back side of bus bar **40**.
(76) Window **42** can be formed to be smaller than electrode terminal **2**. That is, it is not necessary to form a large shape for engaging with electrode terminal **902** as illustrated in FIG. **17**, and it is sufficient if a part of the outer shape and the height can be detected. By making window **42** small through-hole in this manner, the strength of bus bar **40** can be maintained. In other words, according to the present exemplary embodiment, it is possible to eliminate the need for a large hole for inserting the protrusion of electrode terminal **2** as in the related art.
(77) In addition, it is assumed that bus bar **40** is formed larger than electrode terminal **2** in plan view in a state of being overlapped with electrode terminal **2** of battery cell **1**. As described above, even when bus bar **40** is larger than electrode terminal **2** and electrode terminal **2** is inevitably hidden by bus bar **40**, since electrode terminal **2** on the lower side can be visually recognized through window **42** as described above, the welding position and the height can be confirmed, and the reliability of laser welding can be enhanced. However, the above is not essential in the present invention, and for example, even when bus bar **40** is smaller than electrode terminal **2**, the present invention can also be applied to a case where the posture of the part to be welded hidden behind bus bar **40** cannot be confirmed depending on the shape of electrode terminal **2**.
(78) In addition, not only the position of electrode terminal **2** exposed through window **42** but also the height of the front surface can be detected. In the plan view of FIG. **4**, a height detection position where the front surface height of electrode terminal **2** is detected is indicated by HA. FIG. **5** is a schematic cross-sectional view taken along line V-V in FIG. **4**. Since the height of the front surface of bus bar **40** can be detected at a part other than window **42**, the height of the back surface of bus bar **40** can also be calculated by subtracting the known thickness of bus bar **40** from the height of the front surface of bus bar **40**. Accordingly, gap GP between electrode terminal **2** and bus

bar **40** can be calculated from the height of electrode terminal **2** and the height of the back surface of bus bar **40**. At the time of welding bus bar **40** and electrode terminal **2**, it is necessary that there is no gap or it is within a range of a slight gap allowed in welding processing, and this confirmation can be performed. The heights of the front surfaces of bus bar **40** and electrode terminal **2** can be easily determined using a distance measurement sensor. The distance measurement sensor can measure the height of the front surface using a known method capable of measuring the height, such as a triangulation method, a TOF (Time-Of-Flight) type, a capacitance type, an eddy current type, an ultrasonic type, or a contact type.

(79) It is preferable that a plurality of windows **42** are opened in bus bar **40** so that electrode terminal **2** can be visually recognized from each window **42**. With such a configuration, as shown in the cross-sectional view of FIG. **5**, since the different positions of electrode terminal **2** can be confirmed through a plurality of windows **42**A and **42**B, the height of the different parts of electrode terminal **2** can be measured, and the gap GP at the welding position between the back surface of bus bar **40** and electrode terminal **2** can be calculated from the known thickness TB of bus bar **40**. Specifically, in window **42**A on the left in the drawing, distance measurement sensor RF measures height HTR**1** of the upper surface of the battery cell **1**, height HTF**1** of the upper surface of electrode terminal **2**, and height HBF**1** of the upper surface of bus bar **40** forming window **42**A. Then, height HBR**1** on the back surface of bus bar **40** is calculated from thickness TB of bus bar **40** acquired in advance (HBR**1**=HBF**1**−TB). Further, gap GP**1** between the back surface of bus bar **40** and the front surface of electrode terminal **2** in window **42**A can be calculated from height HBR**1** of the back surface side of bus bar **40** and height HTF**1** of the upper surface of electrode terminal **2** (GP**1**=HBR**1**−HTF**1**).

(80) Similarly, also in window **42**B on the right in the drawing, height HTR**2** of the upper surface of battery cell **1**, height HTF**2** of the upper surface of electrode terminal **2**, and height HBF**2** of the upper surface of bus bar **40** forming window **42**B are measured by distance measurement sensor RF, and height HBR**2** on the back surface of bus bar **40** is calculated (HBR**2**=HBF**2**−TB). Further, gap GP**2** between the back surface of bus bar **40** and the front surface of electrode terminal **2** in window **42**B is calculated from height HBR**2** of the back surface of bus bar **40** and height HTF**2** of the upper surface of electrode terminal **2** (GP**2**=HBR**2**−HTF**2**).

(81) Gap GP**0** at the welding position can be calculated from gap GP**1** at window **42**A and gap GP**2** at window **42**B. It is possible to easily calculate the gap of the welding position by designing formation positions of a plurality of windows **42** in advance such that the welding position is located between window **42**A and window **42**B. In the example of FIG. **5**, assuming that the welding position is arranged between window **42**A and window **42**B and the front surface of electrode terminal **2** and the back surface of bus bar **40** are flat, gap GP**0** at the welding position is an average value of gap GP**1** at window **42**A and gap GP**2** at window **42**B.

(82) The plurality of windows **42** are preferably spaced apart and opened so as to sandwich fixing region FA where bus bar **40** and electrode terminal **2** are connected. With such an arrangement, the height of electrode terminal **2** can be measured at two positions facing each other with fixing region FA interposed therebetween, so that it is possible to more accurately calculate the gap in fixing region FA with the gap between electrode terminal **2** and bus bar **40** measured at windows **42** on both sides of fixing region FA. In other words, it is preferable to arrange windows **42** such that a straight line connecting the plurality of windows **42** intersects fixing region FA.

(83) When electrode terminal **2** and bus bar **40** are fixed by laser welding as described above, fixing region FA is a laser welding region. The plurality of windows **42** facing each other with fixing region FA interposed therebetween are preferably opened on a contour defining a connection position connected to electrode terminal **2**. As a result, in addition to the detection of the height, that is, the gap, the confirmation that the electrode terminal **2** is at the connection position can also be performed with the same window **42**.

Second Exemplary Embodiment

(84) The arrangement of windows **42** facing each other is not limited to the example in which the windows are arranged on the left and right sides of the rectangular contour that defines the connection position of electrode terminal **2** as shown in FIG. **4**, and the windows may be arranged on the upper and lower sides, for example. Alternatively, it may be on a diagonal line. Here, as a power supply device according to the second exemplary embodiment, an example of a plan view of connecting piece **41**B of bus bar **40**B including a plurality of windows **42**A′ and **42**B′ opened is illustrated in FIG. **6**. In this example, windows **42**A′ and **42**B′ are opened at diagonal positions of electrode terminal **2**, respectively. As described above, since the height of electrode terminal **2** can be measured at the two diagonal positions, it is possible to more accurately calculate the gap between electrode terminal **2** and bus bar **40**B.

Third Exemplary Embodiment

(85) It is needless to say that the number of the plurality of windows is not limited to two, and may be three or more. For example, in bus bar **40**C of a power supply device according to a third exemplary example illustrated in FIG. **7**, in the contour that defines the connection position of electrode terminal **2** arranged on the back surface of bus bar **40**C, in addition to windows **42**A and **42**B arranged in the middle of the left and right sides, window **42**C is also formed in the middle of the lower side, and positioning and gap confirmation are performed by a total of three windows **42**.

Fourth Exemplary Embodiment

(86) In addition, in bus bar **40**D of a power supply device according to a fourth exemplary embodiment illustrated in FIG. **8**, in addition to windows **42**A and **42**B arranged on the left and right sides of the contour that defines the connection position of electrode terminal **2**, windows **42**C and **42**D are also formed in the middle of the upper and lower sides to form a total of four windows **42**. By providing more windows **42** in this manner, it is possible to increase the number of measurement points and further improve the reliability of connection of electrode terminal **2**.

Fifth Exemplary Embodiment

(87) In addition, in the above example, an example in which the window is mainly opened in a rectangular shape has been described, but in the present invention, the shape of the window is not limited to a rectangular shape, and any shape such as a polygonal shape such as a rhombus shape, a hexagon shape, and an octagon shape, a circular shape, an elliptical shape, and a track shape can be used. In addition, the window does not necessarily need to be formed in an annular shape with its periphery closed, and it is sufficient that the electrode terminal on the lower surface of the bus bar can be visually recognized from above.

(88) For example, in bus bar **40**E of the power supply device according to a fifth exemplary embodiment illustrated in FIG. **9**, window **42**E formed so that an upper left corner part of a contour defining a connection position of electrode terminal **2** can be visually recognized is formed in a notched shape from an end edge of bus bar **40**E. As described above, in the present specification, the window opened in the bus bar means a shape in which a part of the electrode terminal on the back surface of the bus bar can be visually recognized regardless of the name of the window, and is used in a meaning including a notch.

(89) In addition, it is not necessary that all of the plurality of windows are formed on the contour defining the connection position of the electrode terminal arranged on the back surface of the bus bar, and some of the windows may be formed such that only the inside of the window is exposed instead of the contour part of the electrode terminal. Such a window is mainly used for detecting the height of the electrode terminal, that is, the gap between the electrode terminal and the bus bar. For example, in the example of FIG. **9**, lower right window **42**F is used only for detecting the height of electrode terminal **2**, and is not used for confirming the positioning of electrode terminal **2**. As described above, in the present specification, it is not necessary that all the windows can be used to confirm the connection positions of the electrode terminals.

Sixth Exemplary Embodiment

(90) In addition, in the example of FIG. **6** and the like, an example has been described in which the

outer shape of electrode terminal **2** is detected and the height of electrode terminal **2** is performed in each window **42**. However, the present invention is not limited to this configuration, and as the window, an outer-shape window for detecting the outer shape of the electrode terminal and a height window for detecting the height of the electrode terminal may be opened. By separating the windows for outer shape detection and height detection in this manner, the windows can be provided at positions suitable for outer shape detection and at positions suitable for height detection, respectively. In addition, since the outer shape detection and the height detection can be performed in each part, it is possible to obtain an advantage that a plurality of processes can be performed simultaneously in parallel. For example, in bus bar **40**F of the power supply device according to a sixth exemplary embodiment illustrated in FIG. **10**, the outer-shape windows **42-1** are provided diagonally to electrode terminal **2**, while height window **42-2** is provided on both sides so as to sandwich the welding position. By measuring the height position at a part close to the welding position in this manner, it is possible to expect more accurate height detection.

Seventh Exemplary Embodiment

(91) Further, in the above example, the configuration in which the plurality of windows are formed to detect the outer shape and the height of the bus bar has been described, but the number of windows is not limited to a plurality, and only one window may be provided in the present invention. In this case, by forming the window into an elongated shape, the height of the electrode terminal exposed from the window can be detected at a plurality of parts, and the connection position can be confirmed. Such an example is illustrated in FIG. **11** as bus bar **40**G of the power supply device according to a seventh exemplary embodiment. In this example, the window **42**G is formed in a vertically long slit shape so as to vertically straddle the contour defining the connection position. With elongated window **42**G as described above, the height of electrode terminal **2** at different positions can be measured with common window **42**G, and the outer shape of electrode terminal **2** can also be acquired.

(92) In addition, the window preferably has an elongated hole shape longer than the width of the electrode terminal. As a result, the facing end edges of the electrode terminal can be visually recognized from one window, and the inclination of the electrode terminal can be detected from one window. For example, in the example of FIG. **11**, the gap is calculated at each of measurement position MP**7** above window **42**G formed in a vertically long shape and the measurement position MP**8** below the window. Accordingly, the gap between fixing regions FA formed on the left and right sides of measurement positions MP**7**, MP**8** is estimated. In particular, when recess **2***a* is formed in the middle of electrode terminal **2** as shown in FIG. **11**, the height of both sides sandwiching recess **2***a* can be measured to estimate the gap near recess **2***a*. When recess **2***a* is formed in the middle of electrode terminal **2**, recess **2***a* can be recognized as the reference position of the terminal plane.

(93) A plurality of fixing regions for fixing the bus bar and the electrode terminal may be provided as described above. In the example of FIG. **11**, fixing regions FA**1**, FA**2** are arranged on the left and right sides with elongated window **42**G interposed therebetween. As a result, fixing such as laser welding can be performed at a plurality of places in the vicinity of the region where the gap is detected, and the reliability of connection between bus bar **40**G and electrode terminal **2** can be enhanced. In other words, in a case where there is a plurality of fixing regions, it can be said that it is preferable to arrange such that a line connecting the plurality of fixing regions FA**1**, FA**2** intersects window **42**G. Of course, in window **42**G, since bus bar **40**G and electrode terminal **2** cannot be welded, the intersecting part is excluded from the fixing region.

Eighth Exemplary Embodiment

(94) Furthermore, elongated common window **42** is not limited to the configuration in which it is opened so as to straddle the upper and lower sides of the contour that defines the connection position as illustrated in FIG. **11**, and may be arranged so as to straddle the left and right sides of the contour, for example. Alternatively, it may be formed diagonally with respect to the rectangular

contour. An example of this is shown as bus bar **40**H of the power supply device according to an eighth exemplary embodiment in a schematic plan view in FIG. **12**. In this example, window **42**H is opened so as to connect the upper right corner part and the lower left corner part of the rectangular shape defining the contour. Even in such an arrangement, similarly, it is possible to detect the position of electrode terminal **2** and the gap between electrode terminal **2** and bus bar **40**H from common window **42**H, and it is possible to maintain the reliability of the connection between electrode terminal **2** and bus bar **40**H.
(95) The fixing region between the electrode terminal and the bus bar is not necessarily provided at the center of the electrode terminal, and may be provided in a frame shape or a line shape along the outer shape of the electrode terminal, for example. As an example, in the example illustrated in FIG. **12**, fixing regions FA**3**, FA**4** are L-shaped, and L-shaped fixing regions FA**3**, FA**4** are arranged in a posture facing each other along the connection position of electrode terminal **2**.
Method of Manufacturing Power Supply Device
(96) Here, an example of a method of manufacturing the above-described power supply device will be described. First, battery stack **10** in which a plurality of battery cells **1** each including electrode terminal **2** are stacked, and bus bar **40** that connects electrode terminals **2** of adjacent battery cells **1** and includes window **42** opened in advance is prepared. Next, bus bar **40** is overlapped with electrode terminal **2** of battery cell **1**, and a part of electrode terminal **2** is visually recognized through window **42** to confirm whether bus bar **40** and electrode terminal **2** are correctly positioned at the welding position. Then, bus bar **40** whose positioning is confirmed is welded to battery cell **1**. In addition, when the positioning fails, necessary measures are taken such as performing positioning again as necessary. In this way, even in a state where bus bar **40** is stacked on electrode terminal **2**, electrode terminal **2** on the lower side can be visually recognized through window **42**, so that the welding position and the height can be confirmed, and the reliability of welding can be enhanced.
(97) In addition, the step of confirming the positioning can include a step of measuring the height of electrode terminal **2** by visually recognizing a part of electrode terminal **2** through window **42** in a state where bus bar **40** overlaps electrode terminal **2** of battery cell **1**, and a step of calculating the gap between the back surface of bus bar **40** and the front surface of electrode terminal **2** based on the known thickness of bus bar **40**. As a result, by acquiring the gap between the back surface of bus bar **40** and electrode terminal **2**, it is possible to take necessary measures even if the strength of welding between bus bar **40** and electrode terminal **2** cannot be secured when the gap is large, and the reliability of welding can be improved.
(98) Power supply device **100** described above can be used as a power source for a vehicle that supplies electric power to a motor that causes an electric vehicle to travel. As an electric vehicle on which power supply device **100** is mounted, an electric vehicle such as a hybrid automobile or a plug-in hybrid automobile that travels by both an engine and a motor, or an electric automobile that travels only by a motor can be used, and is used as a power source of these vehicles. Note that, in order to obtain power for driving the electric vehicle, an example will be described in which a large number of the above-described power supply devices **100** are connected in series or in parallel, and a large-capacity and high-output power supply device to which a necessary controlling circuit is further added is constructed.
(99) (Power Supply Device for Hybrid Vehicle)
(100) FIG. **13** illustrates an example in which power supply device **100** is mounted on a hybrid automobile that travels by both an engine and a motor. Vehicle HV on which power supply device **100** illustrated in this drawing is mounted includes vehicle body **91**, engine **96** and drive motor **93** that cause vehicle body **91** to travel, wheels **97** driven by engine **96** and drive motor **93**, power supply device **100** that supplies electric power to motor **93**, and generator **94** that charges a battery of power supply device **100**. Power supply device **100** is connected to motor **93** and generator **94** via DC/AC inverter **95**. Vehicle HV travels by both motor **93** and engine **96** while charging and

discharging the battery of power supply device **100**. Motor **93** is driven to cause the vehicle to travel in an area with poor engine efficiency, for example, at the time of acceleration or low speed traveling. Motor **93** is driven by electric power supplied from power supply device **100**. Generator **94** is driven by engine **96** or by regenerative braking when braking the vehicle to charge the battery of power supply device **100**. As shown in FIG. **13**, vehicle HV may be provided with charging plug **98** for charging power supply device **100**. Power supply device **100** can be charged by connecting charging plug **98** to an external power source.

(101) (Power Supply Device for Electric Automobile)

(102) Further, FIG. **14** illustrates an example in which power supply device **100** is mounted on an electric automobile that travels only by a motor. Vehicle EV on which power supply device **100** illustrated in this drawing is mounted includes vehicle body **91**, drive motor **93** that causes vehicle body **91** to travel, wheels **97** driven by motor **93**, power supply device **100** that supplies electric power to motor **93**, and generator **94** that charges a battery of power supply device **100**. Power supply device **100** is connected to motor **93** and generator **94** via DC/AC inverter **95**. Motor **93** is driven by electric power supplied from power supply device **100**. Generator **94** is driven by an energy at the time of regenerative braking of vehicle EV to charge the battery of power supply device **100**. Further, vehicle EV includes charging plug **98**, and power supply device **100** can be charged by connecting charging plug **98** to an external power source.

(103) (Power Supply Device for Power Storage Apparatus)

(104) Furthermore, the present invention does not specify the application of the power supply device as the power source of the motor that causes the vehicle to travel. The power supply device according to the exemplary embodiment can also be used as a power source of a power storage apparatus that charges and stores a battery with electric power generated by solar power generation, wind power generation, or the like. FIG. **15** illustrates a power storage device that charges and stores a battery of power supply device **100** with solar battery **82**.

(105) The power storage device illustrated in FIG. **15** charges the battery of power supply device **100** with electric power generated by solar battery **82** arranged on a roof, a rooftop, or the like of building **81** such as a house or a factory. In this power storage apparatus, the battery of power supply device **100** is charged by charging circuit **83** using solar battery **82** as a charging power source, and then electric power is supplied to load **86** via DC/AC inverter **85**. Therefore, the power storage apparatus has a charge mode and a discharge mode. In the power storage apparatus illustrated in the drawing, DC/AC inverter **85** and charging circuit **83** are connected to power supply device **100** via discharge switch **87** and charge switch **84**, respectively. ON/OFF of discharge switch **87** and charge switch **84** is switched by power supply controller **88** of the power storage apparatus. In the charge mode, power supply controller **88** switches charge switch **84** to ON and switches discharge switch **87** to OFF to permit charging from charging circuit **83** to power supply device **100**. Further, when the charging is completed and the battery is fully charged, or in a state where a capacity larger than or equal to a predetermined value is charged, power supply controller **88** turns off charge switch **84** and turns on discharge switch **87** to switch to the discharge mode, and permits discharging from power supply device **100** to load **86**. As necessary, power supply controller **88** can supply electric power to load **86** and charge power supply device **100** simultaneously by turning on charge switch **84** and turning on discharge switch **87**.

(106) Although not illustrated, the power supply device can also be used as a power source of an electrical storage device that performs power storage by charging a battery using midnight electric power at night. The power supply device charged with the midnight electric power can be charged with midnight electric power that is surplus electric power of a power plant, output electric power in the daytime when an electric power load becomes large, and limit a peak electric power in the daytime to be small. Furthermore, the power supply device can also be used as a power source that charges with both the output of the solar battery and the midnight electric power. This power supply device can efficiently perform power storage effectively using both electric power generated

by the solar battery and the midnight electric power in consideration of weather and electric power consumption.

(107) The power storage system as described above can be suitably used for applications such as a backup power supply device that can be mounted on a rack of a computer server, a backup power supply device for a wireless base station such as a cellular phone, a power source for household or factory power storage, a power source for street lamps, and the like, a power storage apparatus combined with a solar battery, and a backup power source for traffic lights and traffic indicators for roads.

INDUSTRIAL APPLICABILITY

(108) A power supply device according to the present invention, and a vehicle and a power storage apparatus including the power supply device can be suitably used as a power source for a large current used for a power source of a motor for driving an electric vehicle such as a hybrid vehicle, a fuel battery automobile, an electric automobile, or an electric motorcycle. Examples thereof include power supply devices for plug-in hybrid electric automobiles and hybrid electric automobiles capable of switching between an EV traveling mode and an HEV traveling mode, electric automobiles, and the like. Further, the present invention can be appropriately used for applications such as a backup power supply device that can be mounted on a rack of a computer server, a backup power supply device for a wireless base station such as a cellular phone, a power source for power storage for home and factory use, a power source for street lamps, and the like, a power storage apparatus combined with a solar battery, and a backup power source for traffic lights and the like.

REFERENCE MARKS IN THE DRAWINGS

(109) **100**, **900**: power supply device
(110) **1**: battery cell
(111) **1**X: terminal surface
(112) **1***a*: exterior can
(113) **1***b*: sealing plate
(114) **1***c*: gas discharge valve
(115) **2**: electrode terminal
(116) **2***a*: recess
(117) **10**: battery stack
(118) **15**: fastening member
(119) **15***a*: fastening main surface
(120) **15***d*: bent piece
(121) **15***f*: bolt
(122) **16**: insulating spacer
(123) **17**: end surface spacer
(124) **20**: end plate
(125) **30**: insulating sheet
(126) **31**: flat plate
(127) **32**: bent cover part
(128) **40**, **40**B, **40**C, **40**D, **40**E, **40**F, **40**G, **40**H: bus bar
(129) **41**, **41**B: connecting piece
(130) **42**, **42**A, **42**B, **42**A′, **42**B′, **42**C, **42**D, **42**E, **42**F, **42**G, **42**H: window
(131) **42-1**: outer-shape window
(132) **42-2**: height window
(133) **81**: building
(134) **82**: solar battery
(135) **83**: charging circuit
(136) **84**: charge switch

(137) **85**: DC/AC inverter
(138) **86**: load
(139) **87**: discharge switch
(140) **88**: power supply controller
(141) **91**: vehicle body
(142) **93**: motor
(143) **94**: generator
(144) **95**: DC/AC inverter
(145) **96**: engine
(146) **97**: wheel
(147) **98**: charging plug
(148) **901**: battery cell
(149) **802**, **902**: electrode terminal
(150) **903**: end plate
(151) **904**: bind bar
(152) **910**: battery stack
(153) **840**, **940**: bus bar
(154) HA: height detection position
(155) FA, FA**1**, FA**2**, FA**3**, FA**4**: fixing region
(156) GP**0**, GP**1**, GP**2**: gap between electrode terminal and bus bar
(157) RF: distance measurement sensor
(158) HTR**1**, HTR**2**: height of upper surface of battery cell
(159) HTF**1**, HTF: height of upper surface of electrode terminal
(160) HBF**1**, HBF**2**: height of upper surface of bus bar
(161) TB: thickness of bus bar
(162) HBR**1**, HBR**2**: height of back surface of bus bar
(163) HV, EV: vehicle

Claims

1. A power supply device comprising: a battery stack including a plurality of battery cells stacked, the plurality of battery cells including electrode terminals on upper surfaces of outer covering cans of the battery cells, each of the electrode terminals including uppermost surface; and a plurality of bus bars each connecting a corresponding pair of the electrode terminals of adjacent battery cells among the plurality of battery cells, wherein each of the plurality of bus bars is larger than the corresponding one of the electrode terminals of the plurality of battery cells thereby an outline of the corresponding electrode terminal is hidden by the bus bar and defines a window for visually recognizing not all but a part of the uppermost surface of the electrode terminals of at least one of the adjacent battery cells in a state where each of the plurality of bus bars is overlapped with the electrode terminals in plan view in order to detect the outline of the corresponding electrode terminal.
2. The power supply device according to claim 1, wherein each of the plurality of bus bars includes a plurality of windows each being the window opened, and the electrode terminals of the adjacent battery cells are visually recognizable from the plurality of the windows.
3. The power supply device according to claim 2, wherein each of the plurality of bus bars includes the plurality of the windows, the plurality of the windows being separated from each other to sandwich a fixing region where each of the plurality of bus bars and a corresponding one of the electrode terminals is connected.
4. The power supply device according to claim 2, wherein the plurality of the windows are opened at diagonal positions of the electrode terminals.

5. The power supply device according to claim 2, wherein the plurality of windows include: an outer-shape window for detecting an outer shape of each of the electrode terminals; and a height window for detecting a height of each of the electrode terminals.
6. The power supply device according to claim 1, wherein each of the plurality of the windows is in a corresponding one of the plurality of bus bars in a rectangular shape.
7. The power supply device according to claim 1, wherein each of the plurality of the windows is opened in an elongated hole shape longer than a width of the corresponding one of the electrode terminals.
8. The power supply device according to claim 1, wherein each of the plurality of the windows is smaller than the corresponding one of the electrode terminals.
9. The power supply device according to claim 1, wherein a connecting part between each of the plurality of bus bars and the corresponding one of the electrode terminals is a laser welding part.
10. The power supply device according to claim 1, wherein the electrode terminals each includes a recess.
11. A vehicle including the power supply device according to claim 1, the vehicle comprising: the power supply device; a motor for traveling supplied with electric power from the power supply device; a vehicle body equipped with the power supply device and the motor; and a wheel that is driven by the motor to cause the vehicle body to travel.
12. A power storage device including the power supply device according to claim 1, the power storage device comprising: the power supply device; and a power supply controller that controls charging to and discharging from the power supply device, wherein the power supply controller enables charging to the plurality of battery cells with electric power from an outside, and controls charging to the plurality of battery cells.
13. A method of manufacturing a power supply device, the method comprising: preparing a battery stack including a plurality of battery cells stacked, each of the plurality of battery cells including electrode terminals, each of the electrode terminals including uppermost surface, and a bus bar each connecting a corresponding pair of the electrode terminals of adjacent battery cells among the plurality of battery cells, the bus bar being larger than the corresponding one of the electrode terminals of the plurality of battery cells thereby an outline of the corresponding electrode terminal is hidden by the bus bar and defining a window; detecting the outline of the corresponding electrode terminal and confirming positioning to confirm whether the bus bar and the electrode terminals are correctly positioned at a welding position by overlapping the bus bar with the electrode terminals of the plurality of battery cells and visually recognizing not all but a part of the uppermost surface of the electrode terminals through the window; and welding the bus bar whose positioning is confirmed to a corresponding one of the plurality of battery cells.
14. The method of manufacturing the power supply device according to claim 13, wherein the confirming positioning comprises: measuring a height of the electrode terminals by visually recognizing a part of the electrode terminal through the window in a state where the bus bar is overlapped with the electrode terminals of the plurality of battery cells; and calculating a gap between a back surface of the bus bar and front surfaces of the electrode terminals based on a known thickness of the bus bar.
15. The method of manufacturing the power supply device according to claim 13, wherein the welding the bus bar to the corresponding one of the plurality of battery cells is performed by laser welding.